

HOW WOULD THE TEMPERATURES BE NOW IF POLLUTION HAD NOT INCREASED IN 1980?

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INTRODUCTION

To understand the impacts of pollution on temperatures, one must not only examine observed trends but also consider how the climate might have evolved under different historical conditions. One of the most informative approaches in climate science is the construction of counterfactual scenarios, in which researchers model the climate system as it would have behaved without specific human influences. This report investigates one such scenario: what would temperatures today be like if anthropogenic pollution had not begun increasing in 1980? The year 1980 represents a critical turning point in global environmental history, marking the beginning of a period characterized by rapid industrial expansion, accelerated fossil fuel consumption, and significant growth in greenhouse gas emissions. By analyzing this counterfactual world, the report aims to highlight the magnitude of human impact on the climate system and to contextualize present-day warming within broader historical and scientific frameworks.

From the early 1980s onward, global emissions of carbon dioxide, methane, nitrous oxide, and industrial pollutants increased at unprecedented rates. Atmospheric CO₂ concentrations increased from approximately 338 parts per million in 1980 to more than 420 parts per million in recent years, contributing to a measurable increase in global mean surface temperature. Climate datasets and reanalysis products consistently show that the period from 1980 to the present has been one of the fastest-warming intervals in the instrumental record. This warming has been accompanied by more frequent extreme temperature events, shifts in precipitation patterns, and widespread ecological impacts. These observed changes form the baseline against which the counterfactual scenario must be compared.

To estimate what temperatures would be today in the absence of post-1980 pollution, climate scientists rely on model-based reconstructions that simulate the Earth's climate under controlled conditions. These models remove the influence of anthropogenic emissions while preserving natural climate drivers such as volcanic activity, solar variability, and internal ocean-atmosphere dynamics. The resulting simulations provide insight into how the climate system would have evolved if greenhouse gas concentrations had remained near 1980 levels. Although different models vary in their assumptions and computational approaches, they consistently indicate that the majority of warming observed since 1980 is attributable to human activity. In other words, without the rise in pollution that began in that decade, present-day temperatures would be substantially lower.

For this study, we choose the Friuli Venezia Giulia region of Italy as the study region to examine what the temperatures would be like if pollution had not increased in 1980.

In summary, this report explores a scientifically grounded counterfactual scenario to estimate how global temperatures today would differ if pollution had not begun increasing in 1980 for the

Friuli Venezia Giulia region of Italy. To examine this, we work with three models: an Ordinary Least Squares (OLS) regression as the baseline model, a Bayesian regression from the Attribution Climate Impact Model (ATTRICI), and the Conditional Variation Autoencoder model (CVAE).

1 Datasets

1.1 HADCRUT

Hadcrut is a standardized global surface temperature anomaly dataset produced jointly by the Met Office, Hadley Centre, and the Climatic Research Unit (CRU) at the University of East Anglia. It combines land air temperature and sea surface temperature measurements to track historical global temperature changes.

1.2 Precipitation Amount (PP) and Mean Surface Air Temperature (TG)

The precipitation dataset contains daily precipitation totals measured in millimetres (mm), while the mean surface air temperature dataset contains the daily mean temperature recorded in degrees Celsius ($^{\circ}\text{C}$). Both datasets were obtained from the European Climate Assessment & Dataset (ECA&D) / Copernicus C3S.

METHODOLOGY

2 ORDINARY LEAST SQUARES

2.1 Objective and Data

The aim of this part is to build a simple and interpretable baseline model linking global temperature change to local temperature anomalies over a geographical domain covering. The baseline is then used to construct a simplified counterfactual scenario and estimate the local temperature difference between the observed climate and a hypothetical climate without the post-1950 increase in global mean temperature anomaly.

Two datasets are used: HadCRUT5 (monthly global anomalies) and E-OBS (daily European near-surface air temperature). The common analysis period is 1980–2025. E-OBS is restricted to the domain 45° – 47°N and 12° – 14°E .

For HadCRUT5, each monthly global map is reduced to a single global mean temperature anomaly. Grid cells are weighted by latitude:

$$w(\phi) = \cos(\phi),$$

because equal angular grid cells represent smaller physical areas toward the poles. Northern and Southern Hemisphere means are computed separately and combined:

$$GMTA(t) = 0.5 GMTA_N(t) + 0.5 GMTA_S(t).$$

E-OBS daily temperatures are aggregated to monthly means. Local anomalies are computed relative to the 1961–1990 climatology:

$$T_{\text{anom},i}(t) = T_i(t) - T_{i,m}^{1961-1990},$$

where m is the calendar month.

2.2 OLS Model and Smoothing Choice

A simple linear regression is used:

$$T_{\text{anom},i}(t) = a_i + \beta_i \text{GMTA}(t) + \varepsilon_i(t).$$

A value $\beta_i = 1.5$ means that a 1 K increase in GMTA corresponds to a 1.5°C increase in the local anomaly.

Before spatial analysis, the model was tested on the E-OBS grid cell nearest Udine using monthly data and 6-, 12-, and 24-month moving averages. In the first experiment, only GMTA was smoothed; the unsmoothed predictor performed best because, as the document states, “*a very smooth global signal cannot reproduce high-frequency monthly local variability.*”

In the second experiment, predictor and response were smoothed on the same temporal scale. Table 1 shows the results.

Temporal scale	Test R^2	MAE (°C)	RMSE (°C)
Monthly	0.106	1.031	1.276
6 months	0.259	0.531	0.632
12 months	0.371	0.361	0.427
24 months	0.399	0.332	0.379

Table 1: Model performance at different smoothing scales.

Although 24 months performs slightly better numerically, stronger smoothing makes the series mechanically easier to fit. The 12-month moving average is selected as the main specification.

The 12-month model is fitted independently at each valid E-OBS grid cell (at least 80% data availability). The spatial median of β is approximately 1.67°C/K, the mean is 1.61°C/K, and the 5th–95th percentile range is 0.39–2.44°C/K.

The mean spatial in-sample R^2 is approximately 0.463. About 89.2% of valid cells have $R^2 > 0.20$. The threshold is used only for visualization; “*it is not a universal significance threshold.*”[?]

2.3 Counterfactual Scenario and Final Results

A simplified counterfactual is defined by fixing GMTA to its 1951–1960 mean:

$$\text{GMTA}_{\text{cf}} = -0.0806 \text{ K}.$$

For each grid cell, the factual–counterfactual difference is:

$$I_i(t) = \beta_i [\text{GMTA}_{\text{fact}}(t) - \text{GMTA}_{\text{cf}}].$$

The counterfactual anomaly is:

$$T_{\text{cf},i}(t) = T_{\text{fact},i}(t) - I_i(t).$$

The OLS model is a transparent empirical baseline. It uses GMTA as a single predictor and does not explicitly separate volcanic, solar, circulation-related, or other natural sources of variability. The 12-month smoothing reduces short-term fluctuations but does not isolate specific forcings. The resulting sensitivity, fit, and impact maps provide a simple reference for comparison with Bayesian and generative models.

3 ATTRICI MODEL: Bayesian Linear Regression

Understanding how climate change has influenced environmental conditions requires a clear separation between natural climate variability and long-term climate trends. The ATTRICI framework (Attributing Climate Impacts) provides a structured way to construct a counterfactual climate, representing how the climate would have behaved in the absence of long-term warming. This counterfactual climate is then used to quantify the contribution of observed impacts to climate change.

In this analysis, two key datasets are used:

- Daily temperature dataset
- Global mean temperature anomalies

These datasets allow construction of both the factual climate (with observed warming) and the counterfactual climate (with long-term warming removed). ATTRICI removes long-term climate trends correlated with global mean temperature (GMT) while preserving:

- the seasonal cycle,
- day-to-day variability,
- the ranking of each day.

This ensures that the counterfactual climate is stationary, meaning it contains no long-term warming signal but retains realistic variability.

3.1 Deterministic ATTRICI Method

For each day of the year t , the expected temperature is modeled as:

$$g(\mu(T, t)) = C_0(t) + C_1(t)T,$$

where g is the identity link function for temperature.

3.1.1 Seasonal Cycle via One Fourier Harmonic

With only one harmonic, the seasonal structure becomes:

$$C_0(t) = a_0 + a_1 \cos(\omega t) + b_1 \sin(\omega t),$$

$$C_1(t) = c_0 + c_1 \cos(\omega t) + d_1 \sin(\omega t),$$

with:

$$\omega = \frac{2\pi}{365.25}.$$

3.1.2 Factual vs. Counterfactual Expected Values

$$\mu_f(t) = \mu(T(t), t),$$

$$\mu_{cf}(t) = \mu(0, t).$$

3.1.3 Quantile Mapping

1. Factual residual:

$$z_f(t) = \frac{x(t) - \mu_f(t)}{\sigma}.$$

2. Convert to quantile:

$$q(t) = \Phi(z_f(t)).$$

3. Map to counterfactual:

$$z_{cf}(t) = \Phi^{-1}(q(t)).$$

4. Construct counterfactual value:

$$x_{cf}(t) = \mu_{cf}(t) + \sigma z_{cf}(t).$$

3.2 Bayesian ATTRICI Method

The Bayesian version improves stability and provides uncertainty quantification, especially important for regions such as Friuli Venezia Giulia.

3.2.1 Bayesian GLM Structure

The expected value model is:

$$g(\mu(T, t)) = a_0(T) + a_1(T) \cos(\omega t) + b_1(T) \sin(\omega t).$$

The Fourier coefficients depend linearly on GMT:

$$\begin{aligned} a_0(T) &= a_{0,\text{int}} + \beta_0 T, \\ a_1(T) &= a_{1,\text{int}} + a_{1,\text{slope}} T, \\ b_1(T) &= b_{1,\text{int}} + b_{1,\text{slope}} T. \end{aligned}$$

3.2.2 Priors

Gaussian priors:

$$\begin{aligned} a_{0,\text{int}} &\sim \mathcal{N}(15, 10), & \beta_0 &\sim \mathcal{N}(1.4, 0.4), \\ a_{1,\text{int}} &\sim \mathcal{N}(0, 5), & b_{1,\text{int}} &\sim \mathcal{N}(0, 5), \\ a_{1,\text{slope}} &\sim \mathcal{N}(0, 0.15), & b_{1,\text{slope}} &\sim \mathcal{N}(0, 0.15). \end{aligned}$$

Residual variance:

$$\sigma \sim \text{HalfNormal}(3).$$

$$a_{1,\text{slope}} \sim \mathcal{N}(0, 0.1), \quad b_{1,\text{slope}} \sim \mathcal{N}(0, 0.1).$$

Residual variance:

$$\sigma \sim \text{HalfNormal}(1).$$

3.2.3 Posterior Estimation

Posterior distributions for all parameters from the Maximum A Posteriori.

4 Disentanglement of Thermal Anomalies via Physically Constrained CVAE

To decouple the dynamic atmospheric contribution (circulation) from the thermodynamic forcing (global warming) in local temperature anomalies, we formulate a Conditional Variational Autoencoder (CVAE) regularized by physical constraints.

4.1 Data Representation and Physical Prior Initialization

Local daily temperature anomalies ($T_g \in \mathbb{R}^D$) and large-scale atmospheric predictors (sea-level pressure PP and Fourier harmonics of the day of year DOY) are standardized against their daily climatological mean and standard deviation. The thermodynamic forced signal is represented by the standardized global mean temperature anomaly ($fGMT$, computed via a 12-month moving average).

An empirical physical prior derived from observational HadCRUT data is embedded into the network. By evaluating the local covariance between historical temperature records and $fGMT$, we extract an expected baseline sensitivity field ($\alpha_{phys} \in \mathbb{R}^D$, in $^\circ\text{C}/^\circ\text{C}$), which is subsequently mapped into normalized space to initialize the corresponding network parameters.

4.2 Model Architecture and Structural Disentanglement

Signal disentanglement is explicitly enforced by design through the network topology:

- **Encoder:** Takes as input the synoptic weather predictors (PP , DOY) alongside the detached dynamic temperature anomaly, defined as $T_{dyn} = T_g - \alpha \cdot fGMT$. The feature extraction relies on dense layers followed by LayerNorm and SiLU non-linearities, outputting the parameters of a multivariate Gaussian variational posterior: a mean vector μ_z and a log-variance vector $\logvar_z$ (clamped within $[-10, 10]$ for numerical stability). Latent representations z are sampled differentially via the reparameterization trick.
- **Decoder:** The generative process is partitioned into two physically distinct branches:
 1. *Dynamic branch:* Combines z with the weather predictors (PP , DOY) through deep layers to resolve the circulation-driven temperature anomaly μ_{dyn} and the predictive log-variance $\logvar_y$, accounting for residual heteroscedastic uncertainty.
 2. *Thermodynamic branch:* Computes the local climate sensitivity $\alpha \in \mathbb{R}^D$ by modulating a base parameter with seasonal harmonics derived from DOY , constrained through a Softplus activation.

The reconstructed conditional mean is obtained via linear physical superposition: $\mu_{tot} = \mu_{dyn} + \alpha \cdot fGMT$.

4.3 Softplus Constraint and Counterfactual Generation

A Softplus activation ($f(x) = \ln(1 + e^x)$) is applied to the sensitivity field α to guarantee a strictly positive response to global warming ($\alpha > 0$). This constraint prevents physically implausible local cooling trends induced by anthropogenic forcing and regularizes the optimization landscape. Crucially, the base sensitivity parameter is initialized via the inverse-Softplus of the HadCRUT prior, ensuring that training begins from an observationally consistent baseline.

This structural separation enables Pearlian counterfactual inference: by conditioning the decoder on the latent state z inferred from a specific factual event while intervening on the climate forcing ($fGMT = 0$), the model reconstructs the identical synoptic weather event as if it had unfolded in an unperturbed pre-industrial climate.

4.4 Loss Function and Regularization Mechanics

The model is trained end-to-end by minimizing the negative β -ELBO augmented with physical penalty terms:

$$\mathcal{L} = \mathcal{L}_{Recon} + \beta \mathcal{L}_{KL} + \lambda_{mom} \mathcal{L}_{moments} + \lambda_{dyn} \mathcal{L}_{dyn} \quad (1)$$

Under the assumption of diagonal Gaussian likelihood and prior distributions, the reconstruction error and latent KL divergence are evaluated in closed form:

$$\mathcal{L}_{Recon} = \frac{1}{2D} \sum_{j=1}^D \left[\logvar_{y,j} + \frac{(T_{g,j} - \mu_{tot,j})^2}{\exp(\logvar_{y,j})} \right], \quad \mathcal{L}_{KL} = -\frac{1}{2K} \sum_{k=1}^K [1 + \logvar_{z,k} - \mu_{z,k}^2 - e^{\logvar_{z,k}}] \quad (2)$$

The moment-matching penalty $\mathcal{L}_{moments} = (\text{mean}(\alpha) - \mu_\alpha^*)^2 + (\text{std}(\alpha) - \sigma_\alpha^*)^2$ anchors the spatial statistics of the learned sensitivity to the empirical HadCRUT targets.

Conversely, the dynamical penalty $\mathcal{L}_{dyn} = \text{mean}(\mu_{dyn})^2$, originally designed to enforce a zero-mean anomaly constraint on atmospheric circulation, was disabled ($\lambda_{dyn} = 0$). Enforcing weather

dynamics to have a strictly zero mean over short horizons overly penalized the decoder’s capacity to resolve persistent, large-scale asymmetric circulation patterns (e.g., blocking anticyclones and summer heatwaves).

RESULTS

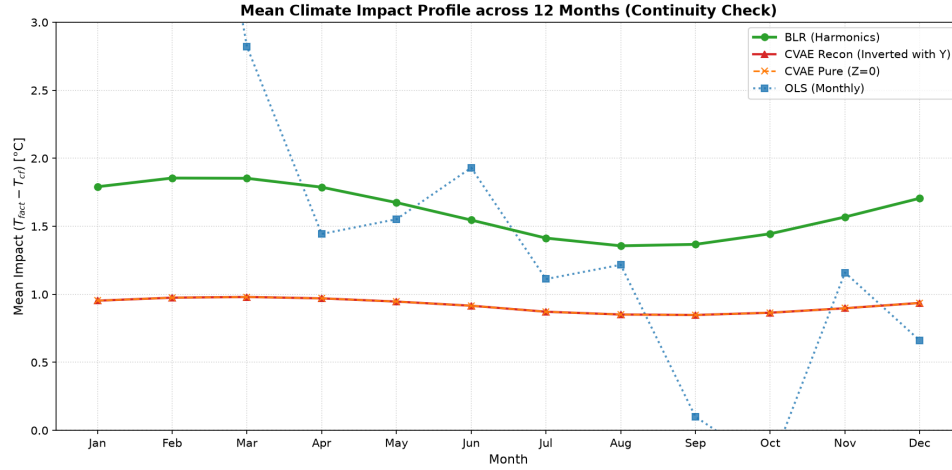


Figure 1: From Figure 1 we can notice that BLR and CVAE have a sinusoidal pattern. That means that the climate impact sensitivity has a seasonal pattern. On the other hand, the OLS model does not capture this trend. The OLS graph shows to include a lot of noise. While the BLR captures a more realistic seasonal cycle the CVAE shows a more stable but smooth pattern cycle.

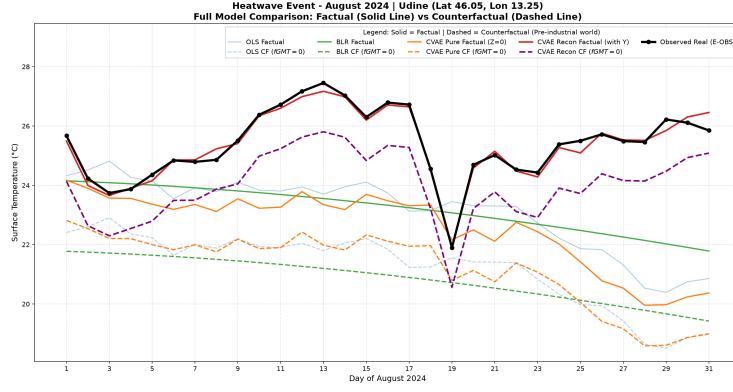


Figure 2: Figure 2 is a graph that measures how much hotter the month of August in Udine for the year 2024. The black line representing the observed data shows an increase in the temperatures above 27°C in the middle of the month then begin to gradually decrease at the end of the month. The counterfactual lines of all the models clearly shows a gap between them and the factual lines indicating how lower the temperatures could have been without the presence of pollution.

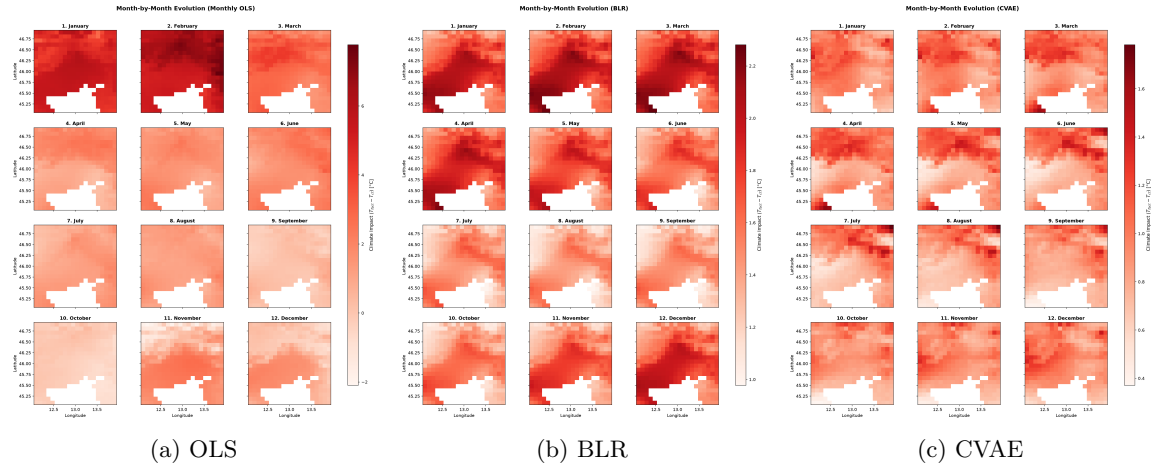


Figure 3: Figure 3 shows the heat map the Friuli Venezia Giulia for all three models. It shows the spatial evidence of climate-change impact during the months of the year. While the BLR and the CVAE show a smooth and coherent heat map over the time, the OLS is filled with noise. The darker red spots occur more during the winter period showing signs of a more stronger warming while the lighter red ones occur during the summer period with signs of not so strong warming

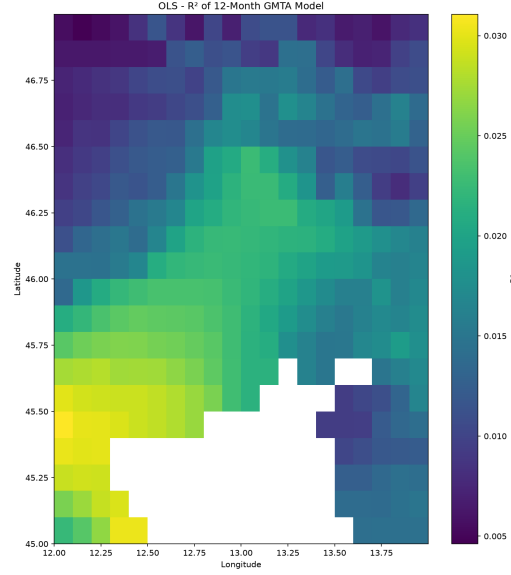


Figure 4: The map shows that the relationship between the 12-month GMTA and local temperature anomalies is relatively weak across Friuli Venezia Giulia. Higher R^2 values are mainly found in the south-western part. This suggests that the ability of global temperature alone to explain local temperature variability changes across the region is generally limited. White areas correspond to grid cells without valid data.

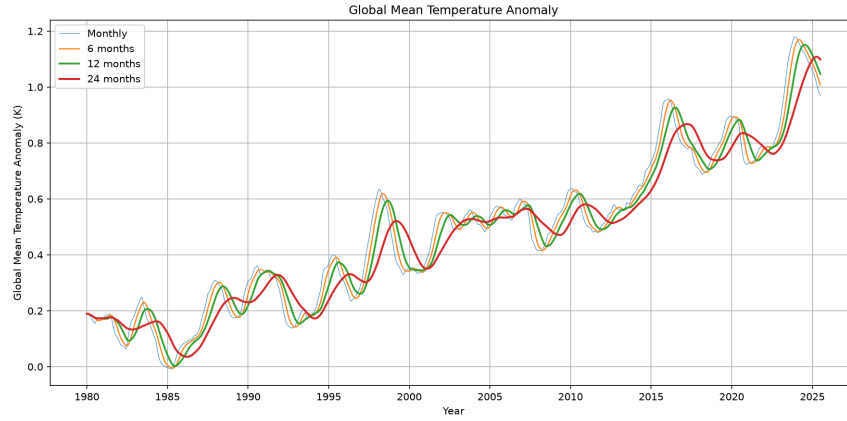


Figure 5: The figure shows a clear long-term increase in global mean temperature anomaly from the 1980s to 2025. The monthly series contains more short-term fluctuations, while the 6-, 12- and 24-month moving averages progressively smooth these variations. The 12-month series provides a good compromise between reducing short-term noise and preserving the main evolution of the global warming signal, which is why it was selected for the main analysis.

Scale	Metric	OLS	BLR	CVAE (Pure)	CVAE (Recon)
Daily	MAE	2.6889	2.4854	2.4886	0.4630
	RMSE	3.3603	3.0890	3.1066	0.6120
	R^2	0.7993	0.8306	0.8299	0.9918
Monthly	MAE	1.5760	1.3263	1.4004	0.2954
	RMSE	2.0176	1.6626	1.7515	0.3954
	R^2	0.9155	0.9426	0.9367	0.9951
Seasonal	MAE	1.1312	0.9238	1.0803	0.2675
	RMSE	1.4135	1.1787	1.3819	0.3496
	R^2	0.9504	0.9650	0.9527	0.9951

Table 2: Model performance comparison across temporal aggregation scales (daily, monthly and seasonal).

5 Bibliography

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